PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Relationship between serum uric acid, all-cause mortality and cardiovascular mortality in peritoneal dialysis patients: systematic review and meta-analysis of cohort studies
AUTHORS	Xue, Xue; Lu, Chun-li; Jin, Xin-yan; Liu, Xue-han; Yang, Min; Wang, Xiao-qin; Cheng, Hong; Yuan, Jun; Liu, Qiang; Zheng, Ruo-xiang; Robinson, Nicola; Liu, Jian-Ping

VERSION 1 – REVIEW

REVIEWER	Choręza, Piotr Slaski Uniwersytet Medyczny w Katowicach, Department of Statistics
REVIEW RETURNED	31-May-2021

GENERAL COMMENTS	The paper is interesting, it should be noted that the authors have a large amount of work in the field of database searches thanks to which the study tinclude a large number of cases (and this is the biggest advantage). Unfortunately, the discusion is superficial, I would expect a critical appraisal of the differences observed in terms of perceiving SUA as a CV mortality predictor between prospective and retrospective studies. I am puzzled by the fact that most of the publications included in the analysis concern China, two more Asian countries (South Korea, Japan) and only one European country. This is a significant limitation of work that, in my opinion, should be clearly definied.

REVIEWER	Chou, Che-Yi China Medical University Hospital, Division of Nephrology
REVIEW RETURNED	13-Jun-2021

GENERAL COMMENTS	The authors analyzed the association of serum uric acid and all-cause, cardiovascular mortality in peritoneal dialysis patients using systematic review and meta-analysis. They found inconsistent results in prospective cohort studies and retrospective cohort studies. This topic is important in clinical and the article is well-written. This article should be considered for publication.

REVIEWER	Cao, Yang Orebro Universitet, School of Medical Sciences
REVIEW RETURNED	14-Jul-2021

GENERAL COMMENTS	In general, the statistical methods used in the study are appropriate. However, regarding nonlinear dose-response analysis using splines, it does not necessarily mean that a linear regression should be considered when P>0.05.

	Besides, I have three minor comments below: 1. Since only cohort studies were included in the current paper, the title of the paper should be revised. 2. The search strategy and detailed results should be provided in supplemental materials. 3. efigure 1 and 2 should be presented in the main manuscript.

VERSION 1 – AUTHOR RESPONSE

Statements to the comments of Reviewer 1:

1. The paper is interesting, it should be noted that the authors have a large amount of work in the field of database searches thanks to which the study tinclude a large number of cases (and this is the biggest advantage).

Unfortunately, the discusion is superficial, I would expect a critical appraisal of the differences observed in terms of perceiving SUA as a CV mortality predictor between prospective and retrospective studies.

Response: Thanks for your helpful comments. We indeed should have a deep discussion in the manuscript.

We found results of the prospective and retrospective cohort studies were inconsistent. we have attempted to discuss the inconsistency of the results from the aspects of participants, exposure, comparability and outcomes. First of all, the prospective cohort study clearly indicated that participants on PD were consecutively enrolled. It is well known that consecutive recruitment is very important to reduce selection bias. While, in some retrospective studies, the process of enrollment was not detailed. The follow-up of the participants was also a prominent issue, including the duration and adequacy of follow-up and the rate of loss to follow-up. In a prospective study, effective measures can be taken to reduce the loss to follow-up rate to avoid bias. The rate of loss to follow-up in a prospective study by Xia X et al. (2016) was only 3.5%, but in some retrospective cohorts, the adequacy and lost follow-up rates were not reported.

Second, hyperuricemia in the PD population was the exposure factor of this study. Both prospective and retrospective cohort studies, the definition of hyperuricemia and the categorization for the SUA level was based on the definition provided in each included article. It should be noted that in retrospective multi-center studies, the measurement methods of SUA may not be uniform across centers. This may lead to measurement bias and have a slight impact on results.

Third, control of the most important and other confounding factors is very important for the comparability between the groups. The most important confounding factors included indicators that can reflect the patient's current residual renal function status. Other confounding factors need to be adjusted and should include gender, age, diabetes history, CVD history, Kt/v, use of UA-lowering drugs, and serum albumin. Most of the studies did not adjust for all potential risk factors. For example, the prospective study of Xia X et al. (2016) lacked adjustment for the confounding factor Kt/v. Therefore, we can not exclude the potential impact of these uncontrolled confounding factors.

Regarding the outcomes, the definition of all-cause and cardiovascular death was clear. However, the handling of patients transferring to hemodialysis therapy, loss to follow-up, and renal transplantation was different for prospective and retrospective studies. The above information from patients was used as censoring data for survival analysis in the prospective study. Whereas, in some retrospective studies, they were directly excluded from the study. This may affect the results and lead to inconsistencies between the prospective and retrospective studies.

Although the risk of bias in the design type of prospective study was relatively small, the interpretation of the results should still be cautious due to the limited quality and quantity of prospective study. Therefore, more high quality studies are needed to clarify the relationship and explore the range of SUA concentration associated with the lowest mortality in the PD patients in future.

2. I am puzzled by the fact that most of the publications included in the analysis concern China, two more Asian countries (South Korea, Japan) and only one European country. This is a significant limitation of work that, in my opinion, should be clearly definied.
Response: We thank the reviewer for pointing out this issue. And we totally agree with the reviewer's opinion. We have added the description in the “Strengths and limitations of this study” section of the paper. (Page4, Line 68-69).
“Of the included studies, eleven were conducted in China, two in other Asian countries and one in Europe, this limits the generalizability of our results.”

Statements to the comments of Reviewer 2:
The authors analyzed the association of serum uric acid and all-cause, cardiovascular mortality in peritoneal dialysis patients using systematic review and meta-analysis. They found inconsistent results in prospective cohort studies and retrospective cohort studies. This topic is important in clinical and the article is well-written. This article should be considered for publication.
Response: Thank you very much for the reviewer's affirmation of our manuscript.

Statements to the comments of Reviewer 3:
1.In general, the statistical methods used in the study are appropriate. However, regarding nonlinear dose-response analysis using splines, it does not necessarily mean that a linear regression should be considered when P>0.05.
Response: Thanks for your helpful comments. We agree with the reviewer's opinion. We have removed the inappropriate description. In general, if there is a dose-response relationship between SUA level and mortality, and P value for non-linear < 0.05, non-linear regression model should be considered. When the P value was close to the critical value of 0.05, both linear and non-linear models needed to be fitted. Since most of the included studies only reported the number of outcomes for the entire cohort population. It was not possible to obtain the number of all-cause and cardiovascular deaths and person-years for each category. We tried our best to contact authors by email or phone in order to acquire the necessary data for the non-linearity test, only one author responded and provided relevant data. A dose-response analysis was not possible. We will continue to focus on this work in future.

2. Since only cohort studies were included in the current paper, the title of the paper should be revised.
Response: Thank you for your suggestion. We have changed the title to “Relationship between serum uric acid, all-cause mortality and cardiovascular mortality in peritoneal dialysis patients: systematic review and meta-analysis of cohort studies.”

3.The search strategy and detailed results should be provided in supplemental materials.
Response: Thanks for your valuable comments. We have added the “the search strategy and detailed results” in supplemental materials.

4. efigure 1 and 2 should be presented in the main manuscript.
Response: We agree with the reviewer's opinion. We have adjusted the image. And efigure 1 and 2 have been presented in the main manuscript.